AVIJIT BARMAN

avijit.barman1598@gmail.com | +91 9064596797 | LinkedIn | GitHub

OBJECTIVE

Aspiring Software Engineer pursuing a Bachelor of Technology (B.Tech) in Computer Science Engineering (Internet of Things) at the Institute of Engineering & Management, Kolkata. Proficient in Java, Python, C, C++, and full-stack web technologies, with hands-on experience in embedded systems, IoT product development, and cross-platform application development. Strong foundation in problem-solving, object-oriented programming, and software development life cycle (SDLC), seeking an internship or entry-level Software Engineer / IoT Developer role to apply technical skills and contribute to real-world engineering solutions.

TECHNICAL SKILLS

- Programming Languages: Java, Python, C, C++
- Web Technologies: HTML, CSS, JavaScript, REST APIs, WebSockets, MQTT
- Frameworks & Libraries: React Native, Node.js, Express.js
- Embedded Systems & IoT: ESP32, Arduino Uno, Arduino IDE, Blynk IoT Platform, Sensor Integration (DS18B20, TDS, HC-SR04 Ultrasonic), Servo Motors, Relay Modules, I2C Communication
- Databases & Cloud: MongoDB Atlas, Cloud-Based Data Storage, API Integration
- Tools & Practices: Git, GitHub, Version Control, Visual Studio, Object-Oriented Programming (OOP), Data Structures, Debugging, Agile Methodology, Problem Solving, Team Collaboration

EDUCATION

Institute of Engineering and Management, Kolkata *2023 – 2027*
Bachelor of Technology in Computer Science Engineering (Internet of Things) — CGPA: 7.85 | West Bengal, India

Sarada Vidya Mandir *2020 – 2022*
Higher Secondary Education — Percentage: 93.4% | West Bengal, India

Sarada Vidya Mandir *2014 – 2020*
Secondary Education — Percentage: 79.28% | West Bengal, India

PROJECTS

Women Safety Smart Watch | ESP32, Arduino IDE, C, Blynk Cloud, NEO-6M GPS *Feb 2026 – Mar 2026*
- Designed and developed an IoT-based wearable women's safety smartwatch by integrating ESP32 microcontroller, NEO-6M GPS, 5 sensors, and 8 hardware modules for 24/7 real-time emergency monitoring with battery-powered, SIM-free operation.
- Implemented 3 emergency detection modes with automatic alert generation, 30-second GPS live location updates, and Blynk Cloud/email notifications by processing sensor data from heart rate, GSR, sound, and motion sensors using embedded C firmware.

AgroMitra | React Native, Node.js, Express.js, MongoDB Atlas, REST APIs *Apr 2026 – Jul 2026*
- Developed a cross-platform full-stack e-Mandi mobile application connecting farmers directly with buyers through real-time crop listing, secure authentication, cloud-based data storage, and transparent agricultural trading.
- Built a scalable full-stack application using React Native, Node.js, Express.js, MongoDB Atlas, and REST APIs, incorporating multilingual support, secure payment workflow, and modular software architecture for future AI-based crop disease detection and price prediction.

FishoTron | Smart IoT Aquarium Management System — ESP32, C++, Blynk IoT *Team Project*
- Collaborated with 3-member team (Team TRIBYTE) to design and build an ESP32-based smart aquarium system for real-time monitoring of water temperature, water quality (TDS), and fish behavior, mentored by a faculty advisor.
- Built a cloud-connected monitoring and control dashboard using the Blynk IoT platform over Wi-Fi, delivering push notifications and remote manual relay control, achieving under 5-second sensor update latency and 95%+ uptime.
- Engineered automated features including scheduled servo-based fish feeding, threshold-triggered automatic water changes (drain/refill relay sequencing), time-based air pump and lighting control via NTP-synced scheduling, and dead-fish detection using an HC-SR04 ultrasonic sensor.
- Integrated a DS18B20 waterproof temperature sensor, analog TDS sensor, and 16x2 I2C LCD for local status display, writing the complete embedded firmware in C++ on the Arduino/ESP32 framework.

CERTIFICATIONS

- Tech Career Skills
- Introduction to Prompt Engineering for Generative AI

ACHIEVEMENTS

- Winner, IEM Quizzophrenia '25 — Technical Quiz Competition conducted by IEM.
- Designed and developed 5+ real-world IoT and embedded systems projects, demonstrating expertise in ESP32, Arduino, sensors, GPS, cloud integration, and automation.

LANGUAGES & INTERESTS

Languages Known: Bengali, English, Hindi

Interests: Database Management Systems, Computer Networks, Internet of Things (IoT)